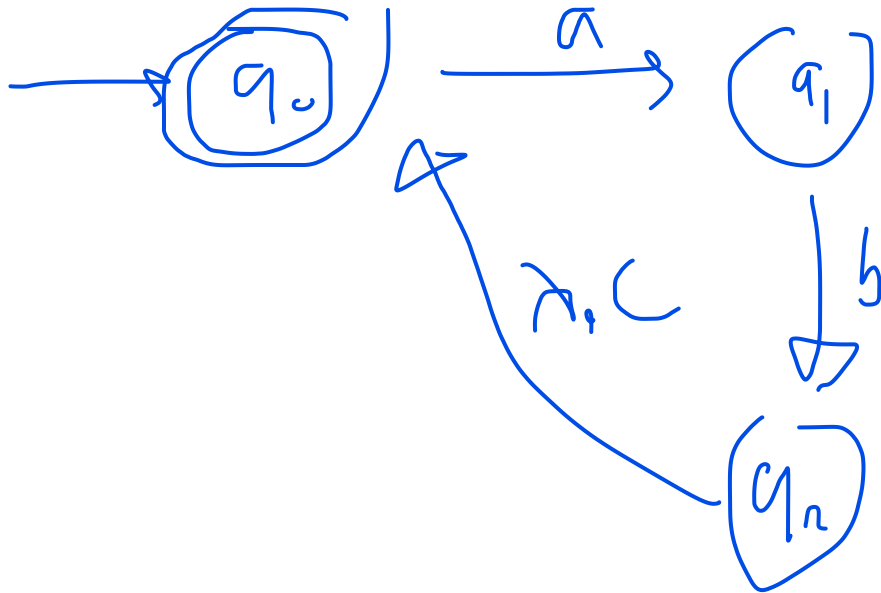


Theory of Computation

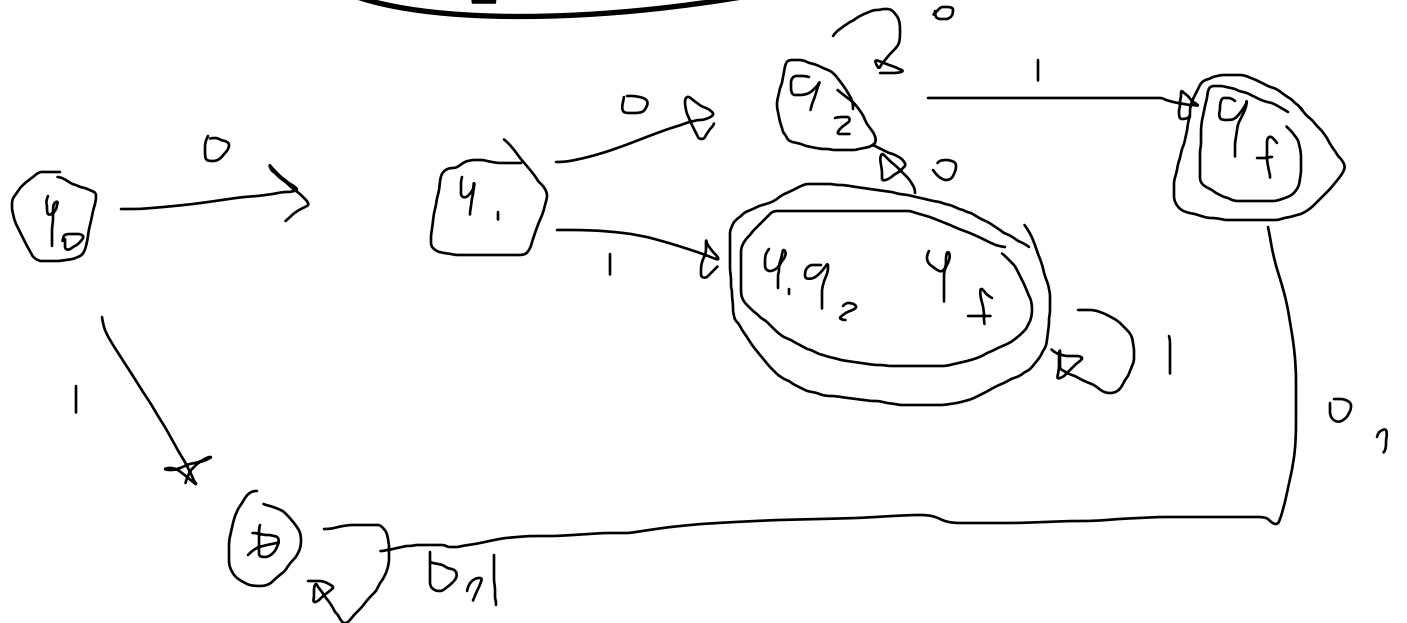
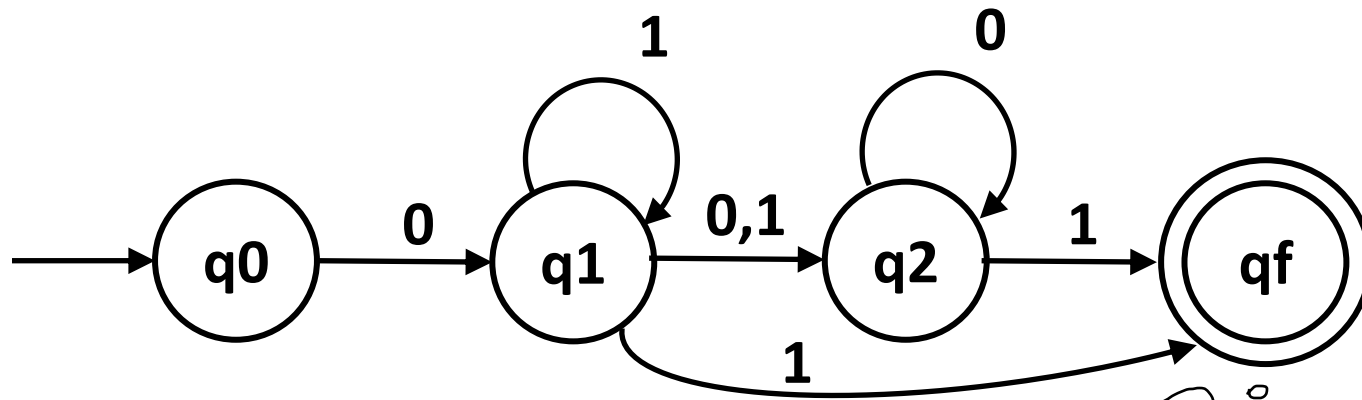
Exercise 3: (Nondeterministic Finite Automata - NFA)

1. Construct the minimal-state NFA that accepts the language $\{ab, abc\}^*$



2. Convert the following NFA to DFA

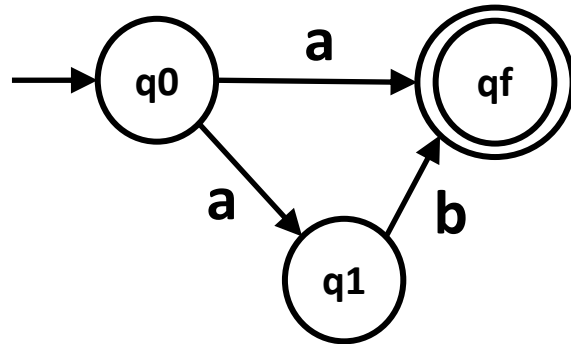
NFA



	0	1
q_0	q_1	$\emptyset$
q_1	q_2	q_1, q_2, q_f
q_2	q_2	q_f
q_f	$\emptyset$	$\emptyset$

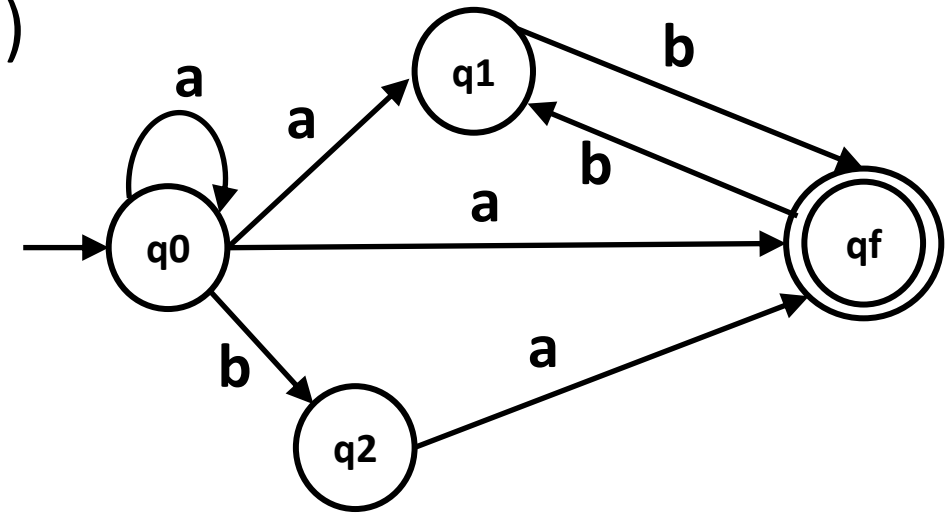
*3. What are the languages accepted by the following NFA ?

a)



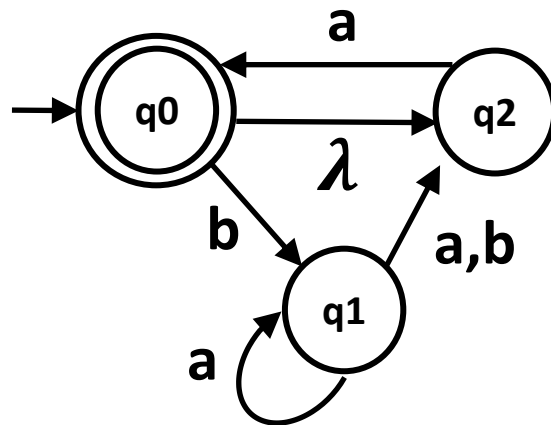
$$L = \{a, ab\}$$

b)



$$L = a^* \cdot \{a, ab, ba\} \cdot \{bb\}^*$$

c)



$$L = \{a, b \cdot \{a\}^* \cdot \{a, b\} \cdot a\}^*$$